

Patryk Drozd

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EDUCATION

MSc, High Performance Computing

Sep 2024 — Sep 2025

Trinity College Dublin

Dublin, Ireland

- A mathematically inclined high performance computing course, with a focus on numerical algorithms and simulation.
- Student Representative for my year group.

Summer School - hpc4climate

28 July 2025 — 7 Aug 2025

ESiWACE & Warm World

Lauenburg, Germany

- Attended a [summer school](#) involved with working on and learning about climate science within the context of heterogeneous computing and data science.

BSc, Theoretical Physics and Mathematics

Sep 2021 — Sep 2024

Maynooth University

Maynooth, Co. Kildare, Ireland

- A 3 year accelerated theoretical physics and pure mathematics course.
- Student representative for my year group in Physics department.

EXPERIENCE

Teaching Assistant

Sep 2024 — May 2024

Trinity College Dublin

Dublin, Ireland

- Teaching assistant for a first year engineering mathematics module involving teaching and grading students assignments.

Research Intern

June 2024 — Sep 2024

Fraunhofer IWES

Oldenburg, Lower Saxony, Germany

- Developed benchmarking scripts for a particular site with data masts and wind turbines. This involved developing skills in data analysis through Pandas and computational fluid dynamics with OpenFOAM.

Undergraduate Researcher

June 2023 — Aug 2023

Maynooth University

Maynooth, Co. Kildare, Ireland

- I worked with a postdoctoral researcher to develop skills in machine learning with Python and Pytorch. I developed models to classify regions of space for whether they form features like black holes or stars, and to upscale coarse snapshots from cosmological simulations. A [link](#) the research poster.

SKILLS

- **Programming**
 - Python — Proficient with libraries such as Numpy, Matplotlib, Pandas and Pytorch.
 - C/C++ — Proficient with ability to write multi threaded code with OMP, MPI, and CUDA.
 - Linux — Using machines including university HPC systems and WSL with proficiency in Bash scripting.
- **Languages**
 - Fluent in English and Polish. Basic German skills.

AWARDS

- **Gold MUSE Award** *Maynooth University*
 - An award recognising student contribution to non credit bearing activities.

PROJECTS

GPU Delaunay Triangulation

- For my masters project I'm currently working on implementing a Delaunay triangulation meshing algorithm on the GPU with C++ and CUDA.

Finite Element PDE solvers

- As part of a significant undergrad project I worked on Finite Element solvers which numerically solved various linear PDEs using Python and Numpy. Some [pictures](#).